

Lab Title:- ARP POISONING FORENSICS

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Course Name:- OWASP TOP 10 Vulnerabilities and Exploitation Techniques

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Executive summary

The covers a practical understanding of a normal ARP resolution and a arp poisoning covering some key aspect such as arp response/reply, normal broad cast.

Lab Objectives

The objectives of the lab is to understand the difference between a what is a normal address resolution protocol(ARP) resolution and a ARP poisoning using a captured ARP traffic.

Tools and Resources Used

- Kali linux:- A penetration testing platform used for security assessments.

Kali linux tools

- tshark:- Dump and analyze network traffic
 - wireshark:- is a GUI network protocol analyzer that lets you interactively browse packet data from a live network or from a previously saved capture file.
 - sha256sum:- Print/compute or check SHA256 (256-bit) checksums message digest.
 - Python 3 scapy:- Interactive packet manipulation tool
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Methology

- I launched the kali linux terminal.
- I created a directory and then enter in to the directory
- I used the w get command to clone/download arp traffic file(arp.pcap)
- I made a copy of the smtp.pcap file and the compare it hash value with the original arp.pcap file using shasum256 command
- Using different tshark command option to analyse the arp.pcap file based on some aspect such as arp response/reply, normal broad cast

Screenshot and Evidence

SCREENSHOT SHOWING THE INTERFACE, ROUTE & ARP TABLE OF THE CAPTURED ARP TRAFFIC

```
(kali㉿kali)-[~/SBT-DF203-Lab5]
$ ip -br address | tee reports/interfaces.txt
lo                UNKNOWN          127.0.0.1/8 ::1/128
eth0              UP                192.168.152.129/24 fe80::61f5:60f0:660e:d7c6/64

(kali㉿kali)-[~/SBT-DF203-Lab5]
$ ip route | tee reports/routes.txt
default via 192.168.152.2 dev eth0 proto dhcp src 192.168.152.129 metric 100
192.168.152.0/24 dev eth0 proto kernel scope link src 192.168.152.129 metric 100

(kali㉿kali)-[~/SBT-DF203-Lab5]
$ ip neigh show | tee reports/arp_table_initial.txt
192.168.152.2 dev eth0 lladdr 00:50:56:ed:40:31 STALE
```

SCREENSHOT SHOWING THE NORMAL BROADCAST ARP REQUEST OF THE CAPTURED ARP TRAFFIC

```
(root㉿kali)-[/home/kali/Music]
# tshark -r evidence/arp.pcap -Y 'arp' -T fields \
-e frame.number -e frame.time -e eth.src -e eth.dst -e arp.opcode \
-e arp.src.proto_ipv4 -e arp.src.hw_mac -e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
| tee reports/normal_arp_fields.tsv

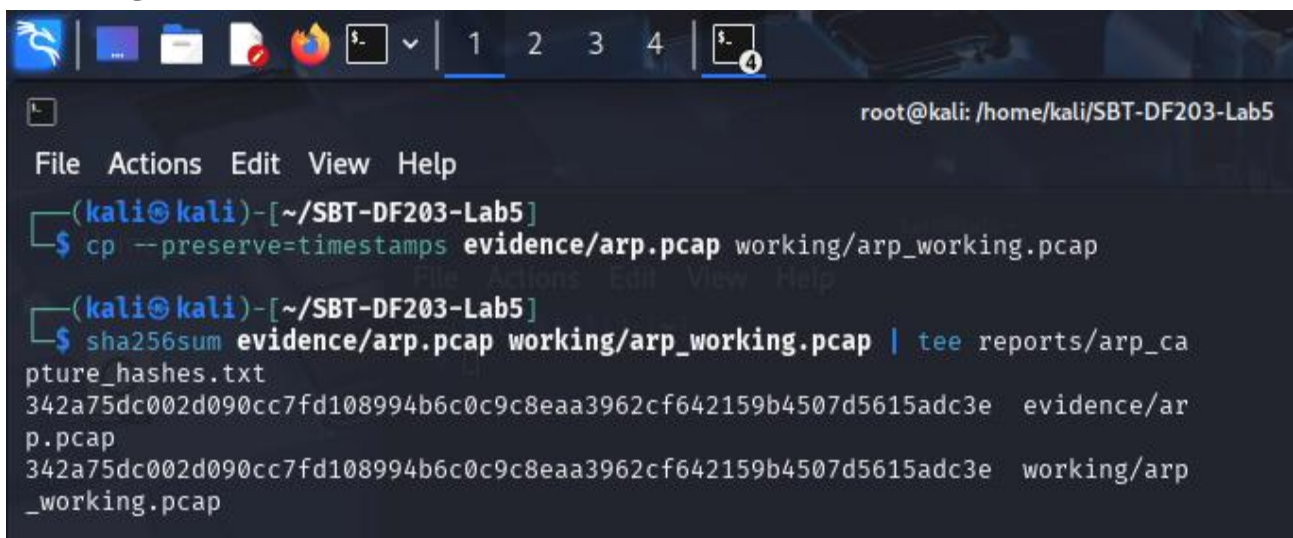
Running as user "root" and group "root". This could be dangerous.
1   Mar 11, 2023 22:22:03.059513401 EST    00:1b:17:00:0a:30    ff:ff:ff:ff:ff:ff    1    136.160.215.1  00:1b:17:00:0a:30    1
36.160.215.1    ff:ff:ff:ff:ff:ff
2   Mar 11, 2023 22:22:38.192015486 EST    00:1b:17:00:0a:30    ff:ff:ff:ff:ff:ff    1    136.160.215.1  00:1b:17:00:0a:30    1
36.160.215.199 ff:ff:ff:ff:ff:ff
3   Mar 11, 2023 22:22:44.903581555 EST    00:50:56:86:cb:fc    ff:ff:ff:ff:ff:ff    1    136.160.215.15 00:50:56:86:cb:fc    1
36.160.215.194 00:00:00:00:00:00
4   Mar 11, 2023 22:22:44.904123642 EST    00:50:56:86:02:65    00:50:56:86:cb:fc    2    136.160.215.194 00:50:56:86:02:65    1
36.160.215.15  00:50:56:86:cb:fc
5   Mar 11, 2023 22:22:49.943023192 EST    00:50:56:86:02:65    00:50:56:86:cb:fc    1    136.160.215.194 00:50:56:86:02:65    1
36.160.215.15  00:00:00:00:00:00
6   Mar 11, 2023 22:22:49.943043799 EST    00:50:56:86:cb:fc    00:50:56:86:02:65    2    136.160.215.15  00:50:56:86:cb:fc    1
36.160.215.194 00:50:56:86:02:65
7   Mar 11, 2023 22:23:03.059262685 EST    00:1b:17:00:0a:30    ff:ff:ff:ff:ff:ff    1    136.160.215.1  00:1b:17:00:0a:30    1
36.160.215.1    ff:ff:ff:ff:ff:ff
8   Mar 11, 2023 22:23:06.191965551 EST    00:1b:17:00:0a:30    ff:ff:ff:ff:ff:ff    1    136.160.215.1  00:1b:17:00:0a:30    1
36.160.215.183 ff:ff:ff:ff:ff:ff

(root㉿kali)-[/home/kali/Music]
```

SCREENSHOT SHOWING NORMAL ARP REPLY OF THE CAPTURED ARP TRAFFIC

```
root@kali: ~/home/kali/GBT-DF203-Lab5
# tshark -r working/arp_working.pcap -Y 'arp.opcode=2' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst \
-e arp.src.proto_ipv4 -e arp.src.hw_mac -e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
| tee reports/arp_replies.tsv
Running as user "root" and group "root". This could be dangerous.
4 1678591364.904123642 00:50:56:86:02:65 00:50:56:86:cb:fc 136.160.215.194 00:50:56:86:02:65 136.160.
215.15 00:50:56:86:cb:fc
6 1678591369.943043799 00:50:56:86:cb:fc 00:50:56:86:02:65 136.160.215.15 00:50:56:86:cb:fc 136.160.
215.194 00:50:56:86:02:65
```

SCREENSHOT SHOWING THE SUPPLIED HASH VALUE OF THE CAPTURED ARP TRAFFIC



```
root@kali: ~/home/kali/GBT-DF203-Lab5
File Actions Edit View Help
(kali@kali)-[~/GBT-DF203-Lab5]
$ cp --preserve=timestamps evidence/arp.pcap working/arp_working.pcap
(kali@kali)-[~/GBT-DF203-Lab5]
$ sha256sum evidence/arp.pcap working/arp_working.pcap | tee reports/arp_capture_hashes.txt
342a75dc002d090cc7fd108994b6c0c9c8eaa3962cf642159b4507d5615adc3e evidence/arp.pcap
342a75dc002d090cc7fd108994b6c0c9c8eaa3962cf642159b4507d5615adc3e working/arp_working.pcap
```

SCREENSHOT SHOWING ARP INVENTORY OF THE CAPTURED ARP TRAFFIC

```
root@kali: ~/home/kali/GBT-DF203-Lab5
# tshark -r working/arp_working.pcap -Y 'arp.opcode=2' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst \
-e arp.src.proto_ipv4 -e arp.src.hw_mac -e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
| tee reports/arp_replies.tsv
Running as user "root" and group "root". This could be dangerous.
4 1678591364.904123642 00:50:56:86:02:65 00:50:56:86:cb:fc 136.160.215.194 00:50:56:86:02:65 136.160.
215.15 00:50:56:86:cb:fc
6 1678591369.943043799 00:50:56:86:cb:fc 00:50:56:86:02:65 136.160.215.15 00:50:56:86:cb:fc 136.160.
215.194 00:50:56:86:02:65
```

SCREENSHOT SHOWING IP-TO-MAC CLAIM SUMMARY OF THE CAPTURED ARP TRAFFIC

```
(root@kali)-[/home/kali/SBT-DF203-Lab5]
# tshark -r working/arp_working.pcap -Y 'arp.opcode==2' -T fields -e arp.src.proto_ipv4 -e arp.src.hw_mac \
| sort | uniq -c | sort -nr | tee reports/ip_mac_claims.txt
Running as user "root" and group "root". This could be dangerous.
  1 136.160.215.194 00:50:56:86:02:65
  1 136.160.215.15  00:50:56:86:cb:fc

(root@kali)-[/home/kali/SBT-DF203-Lab5]
```

SCREENSHOT SHOWING CONFLICTING GATEWAY MAC EVIDENCE OF THE CAPTURED ARP TRAFFIC

```
(root@kali)-[/home/kali/SBT-DF203-Lab5]
# tshark -r working/arp_working.pcap -Y 'arp.opcode==2 && eth.dst!=ff:ff:ff:ff:ff:ff' -T fields \
-e frame.number -e frame.time -e arp.src.proto_ipv4 -e arp.src.hw_mac -e eth.dst \
| tee reports/unicast_arp_replies.tsv
Running as user "root" and group "root". This could be dangerous.
  4   Mar 11, 2023 22:22:44.904123642 EST    136.160.215.194 00:50:56:86:02:65    00:50:56:86:cb:fc
  6   Mar 11, 2023 22:22:49.943043799 EST    136.160.215.15  00:50:56:86:cb:fc    00:50:56:86:02:65

(root@kali)-[/home/kali/SBT-DF203-Lab5]
```

SCREENSHOT SHOWING THE RESTORED ARP TABLE AND PROCESS CHECK

```
(root@kali)-[/home/kali/SBT-DF203-Lab5]
# sudo ip neigh flush all

(root@kali)-[/home/kali/SBT-DF203-Lab5]
# ip neigh show
192.168.200.1 dev eth0 INCOMPLETE

(root@kali)-[/home/kali/SBT-DF203-Lab5]
# ps aux | grep -E '[a]rp.py|[s]capy' | tee reports/process_check.txt

(root@kali)-[/home/kali/SBT-DF203-Lab5]
```

Conclusion

The lab has helped me in understanding the difference between a normal ARP Resolution and an ARP poisoning.

Reference

➤ICDFA course Material (SBT-DF203(basic networking skills for digital forensics)- by
COMMANDANT AMINU IDRIS

➤ICDFA Recorded sessions
